SIDDHARTHA ACADEMY OF HIGHER EDUCATION

(Deemed to be University)

VELAGAPUDI RAMAKRISHNA SIDDHARTHA SCHOOL OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech III Year, ACADEMIC YEAR: 2026-27

24CS381-ENGINEERING PROJECTS FOR COMMUNITY SERVICES(EPICS)
B.Tech (Computer Science & Engineering) - AI&ML

Review 0: Template

Course: 24CS381 – Engineering Projects for Community Services (EPICS)

Section: CSE-AI & ML Batch. No:

Name of the Students and Roll. No along with CGPA:

S.No	Name of the Student	Roll. No	CGPA
1.	M. Kavya	24EU02030	8.67
2.	N. Bhagya sri	24EU02038	8.10
3.	M. Harini	24EU02029	7.98

Title of the project:
AI-Assisted Early Grain Spoilage Prediction and Smart Storage Safety Monitoring System

Societal Relevance: Health/Agriculture/Education/Village administrationetc..,
Agriculture / Food Safety / Smart Storage Monitoring

Client Details and Location (Societal) with Photograph (paste here):

Client Name: M. Murali Mohan
Business: Rice Mill and Grain Storage Unit
Location: Gudivada, Andhra Pradesh

The client is associated with rice mill and grain storage operations. Improper storage conditions may lead to moisture issues, fungal growth, and grain spoilage. The proposed system helps monitor storage conditions and provide early spoilage alerts.

Abstract:

Grain spoilage is a major problem in rice mills and grain storage units due to excess moisture, temperature changes, and poor storage conditions. Existing systems mainly depend on manual monitoring and basic environmental checking, which may identify spoilage only after damage occurs.

This project proposes an AI-Assisted Early Grain Spoilage Prediction and Smart Storage Safety Monitoring System using IoT sensors and AI-based analysis. The system continuously monitors temperature, humidity, and gas conditions inside storage areas and provides early alerts when spoilage risk is detected. This helps reduce food wastage, improve grain safety, and support storage operators.

Keywords

- AI
- IoT
- Grain Monitoring
- Smart Storage
- Spoilage Prediction
- Agriculture
- Sensors
- ESP32

Signature of the EPICS Guide